Automated System for Prostate Cancer Detection from Histopathology Images Using XGBoost

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Abstract. Prostate cancer is a cancer that is common in men in the world and proper early detection is the drive to treatment. A machine learning architecture of classifying prostate histopathology images on the basis of SICAPv2 dataset, with RGB tissue patches and masks of the position of the cancer, is presented. The suggested algorithm is XGBoost with handwritten texture attributes, such as GLCM, LBP, and HOG. Very elaborate deep learning framework like MobileNetV2 (dual-head) and ResNet50 were also tested in comparison. Of all the experiments, the XGBoost model with both GLCM+LBP features recorded the highest scores with an accuracy of 84%, and ROC-AUC of 92%. The results demonstrate that texture-based machine learning techniques can be better in comparison with deep networks in case of a small-sized dataset, which may be considered a reliable and computationally effective approach to supporting prostate cancer detection and further clinical decision-making.

Keywords: AI-Learning · Deep Learning· Histopathology· Machine Learning· Prostate Cancer· Whole-Slide Images

1 Introduction

One of the most prevalent cancers among men is the prostate cancer which is a serious health issue in the global society. It begins in the prostate gland that is found beneath the bladder. The cancer normally proceeds on slowly, however it is very important to detect it at a young age so that one can determine whether a patient requires effective treatment measures such as surgery or radiations. Douglass [1] reported 1.47 million cases of newly diagnosed prostate cancer in the year 2022, 7.3 percent of the total cancer cases. It was the fourth-ranking malignancy throughout the earth and the second ranking among males, having 14.2% of cancer cases in males. Prostate cancer was the death cause among

397,430 or 4.1 percent of all cancer deaths worldwide which is the number eight of all cancer deaths. More than 5 million men who had this cancer had survived five years after diagnosis. Physicians normally begin by testing PSA levels and carrying out Digital Rectal Examination (DRE). Deep learning is more rapid, it is reliable, and it is able to detect certain patterns of those images that could be overlooked by humans. Singhal et al. [2] created a deep-learning architecture via frame-based style, trained on data of vast regions, through active learning, and an enhanced U-Net architecture. The model was accurate on data in their own hospital (89.4%) and other locations (83.1%). This was in accordance with the scores of expert pathologists. Nevertheless, the research indicates that the model has to be compared with long-term patient outcomes to demonstrate that it can be used to predict prognosis precisely.

Problem Statement:

Visual examination of biopsy slides is a crucial element in the diagnosis of the prostate cancer, which is not only time-consuming but it is also subjective. Pathologists will also give varied results when looking at the same tissue sample resulting in the issuance of inconsistent grades and even different treatment choices. This inconsistency shows that there is urgent demand to come up with objective and consistent computational techniques that can help in supporting and standardizing the process of prostate biopsy evaluation. People have found deep learning models perform well in medical image analysis, however, they usually demand large well-annotated scale, which is scarce in prostate histopathology. Even the literature does not cover much on whether texture-based machine learning procedures can compete better than deep networks in the case of small data. This gap inspires the necessity to explore lightweight, interpretable frameworks with the capability of providing high precision without a large amount of training data.

The objectives are:

-To analyze the prostate histopathology images into cancerous and non-cancerous categories.

-To assess the model performance in the terms of accuracy, sensitivity, specificity and ROC-AUC on independent test sets.

2 Literature Survey

Deep learning, in particular machine learning is transforming how physicians approach cancer tissues. Such procedures are able to detect and classify cancer accurately as professionals.

Prostate cancer diagnosis using Machine Learning: Federated learning is primarily a distributed machine learning paradigm.It is a specialized training methodology which is a part of ML. Fedrated learning trains a shared model across multiple independent local datasets .It is used to overcome large-scale, diverse, and well-annotated data. The works of federated learning in the cancer care setting have already been reviewed by Nasajpour et al. [3]. The point is to create a global model based on the data of numerous hospitals without the

transportation of data. Lu et al., [4] looked at federated learning in medical image segmentation. They classified methods based on the speed at which they convey information, the degree to which they preserve privacy and the nature of diversified information. Although the method allows hospitals to work together, communication is expensive and the poor results with non-IID information still pose a setback with the new system. The future work should develop less expensive protocols to communicate. Li [5] introduced label-efficient self-supervised learning as well as few-shot segmentation-based prostate imaging in this manner. Jahanifar et al., [6] investigated pathology domain-generalization and discovered many instances and found out that simple stain-augmentation ranks high against more complex algorithms.

Prostate cancer diagnosis using Deep Learning: DL uses Artificial Neural Networks (ANNs) with multiple hidden layers to learn feature representations automatically from large amount of raw data . A deep-learning system, which Singhal et al., had developed [2] shows presence of the cancer. The system is programmed to accept any type of image of tissues and has a special network, which learns what aspects of the image are important. It achieved 89.4 percentage in same hospital data and 83.1 percentage in non-same hospital data, which is nearly equivalent to the expert doctors. It is able to display heatmaps that can explain its decisions, but it can be as well deceived by bad tissue or it is not yet tested over long periods of time. Kondejkar et al., [7] developed a multi-scale deep-learning framework to grade prostate cancer based on DiagSet data set that consists of nine grades. They took ready-prepared ResNet networks and discovered that ResNet 34 provided nearly ideal output at varying zoom levels.

Prostate cancer diagnosis using ViTs and Multimodal: Vision Transformers (ViTs) are the advanced Deep Learning architecture that applies the Transformer model.They have demonstrated best performance, specially in tasks which require an understanding of global context , large-scale image classification and segmentation problems. Khan et al., [8] reviewed Deep-Vision transformers (ViTs) in medical imaging. ViTs have the ability to learn long-term connections but require much pre-training, which restricts their applicability to small data sets. It was proposed to provide a balance between power and speed. For prostate cancer, particularly, recent reviews and technical papers [9], [10], [11] and [12] report the success of multimodal machine learning/deep learning fusion, attention-based architectures, and neural network diagnostic strategies .

Other related works: AI-assisted diagnostic pipelines are mentioned as one of the main outcomes of recent research in computational histopathology. The increasing influence of deep learning and transformer-based techniques in cancer detection and processing has been pointed out by extensive reviews [13] and [14]. Research on mpMRI-based segmentation and classification that supports the above work, has also been reporting that the use of deep learning techniques can significantly increase the performance of prostate cancer detection [15].

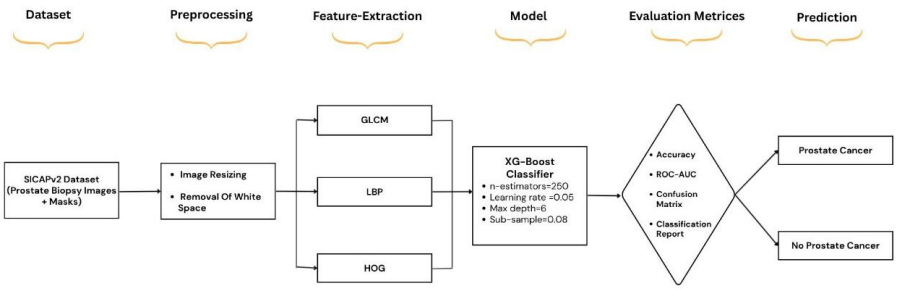


Fig. 1: Methodology Flow Diagram

The above figure reflects the suggested prostate cancer detection pipeline that consists of preprocessing and irrelevant regions removal, texture features (HOG, GLCM, and LBP) extraction, data equalization, and end-of-pipe classification (the XGBoost ensemble). The chosen stages maximise the feature discriminability and classification strength with merging complementary handcrafted features with a powerful ensemble learning algorithm.

3 METHODS AND MATERIALS

Dataset Preparation: The SICAPv2 dataset provides RGB prostate tissue images paired with masks that highlight the cancerous regions (1) and non-cancerous regions (0). The pairing of each image with its corresponding mask through filename ensures dependable supervision throughout the training of the model. The dataset consists of 18,783 pairs of images and masks, which were all obtained from the annotated whole-slide images (WSIs). The original WSI is cut into non-overlapping square patches of 256 by 256 pixels and for every patch the corresponding binary mask which shows the presence or the absence of cancerous tissue is created. The distribution of classes in the dataset is 8,063 positive patches (lesion present) and 10,720 negative patches (no lesion), which gives a balanced representation for training and testing the model as shown in fig 1.

Preprocessing: Downsizing all the images to 256 x 256 pixels. Each image is divided into 32 x 32 block tiles which has crossed 50%. A threshold of 5% for tissue was applied to eliminate the tiles with background-dominant regions, as these areas contribute very little to the diagnostics. Around 5,728,453 tiles were discarded from 18,783 images that produced 1,024 tiles each (totaling 19,233,792 tiles), and the major part of the eliminated tiles consisted of non-informative whitespace.

Feature Extraction: Three key feature sets are extracted, including:

- Gray-Level Co-Occurrence Matrix
- Local Binary Pattern
- HSV histogram

1. **Gray -Level Co-occurrence Matrix (GLCM)**

GLCM determines second-order statistics of a texture (contrast, homogeneity, energy, correlation) through the analysis of the relationships between pixel intensities as a spatial relationship. The tissue structure in terms of distances 1 and 2, and four orientations $(0, \pi/4, \pi/2, 3\pi/4)$ were used to extract features.

2. **Local Binary Pattern (LBP)**
   LBP coded micro-texture by thresholding neighbours around cells to reveal books at the cell-level, boundaries around the nucleus as well as any structural anomalies. Histograms of uniform LBP were calculated using radius = 1 and 8 neighbors, which create a small and noise-resistant feature which is suitable to use with XGBoost.
3. **Histogram of Oriented Gradients (HOG)**
   HOG is used to extract macro-structural data based on edge orientation data and gradient flow data in tissue aspects. The HOG descriptors of the 256x256 tiles were produced with 9 orientations, 16x16 cell pixels and 2x2 cell blocks to result in features of structural abnormalities indicative of cancer.
   Despite being generic descriptors, GLCM, LBP, and HOG are still useful for capturing structural patterns, texture, and edges in histopathology tiles. Even though they don't directly address stain variability or complex glandular morphology, these features offer helpful cues that aid the model in learning discriminative patterns for lesion prediction.

Classification based on Machine Learning: The proposed system utilizes XGBoost and MobileNetV2 for classification.

1. **XGBoost Classifier**

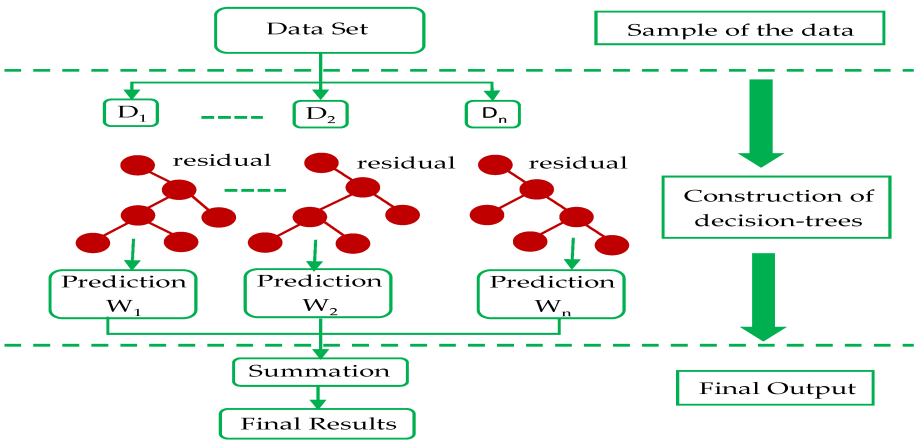


Fig. 2: Architecture of XGBoost

XGBoost constructs trees in a sequential fashion, fixing the mistakes of earlier trees. It uses gradient and second-order (Hessian) information to choose the best splits, applies strong regularization (L1/L2) to avoid overfitting, and uses

shrinkage and subsampling to improve generalization as shown in Fig 2. The final prediction is the combined output of all trees, making XGBoost fast, accurate, and highly effective for structured feature data like GLCM, LBP, and HOG.

MobileNetV2

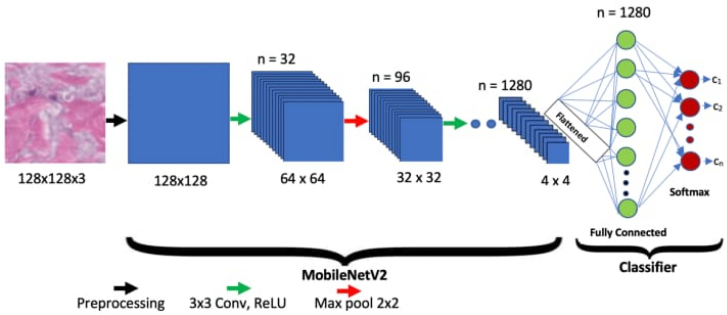


Fig. 3: Architecture of MobileNetV2

MobileNetV2 starts by applying depthwise and pointwise convolutions, which reduces the amount of computation while maintaining the significant patterns. These features are successively downsampled and enlarged via inverted residual blocks as shown in Fig 3. In the end, a fully connected layer along with a softmax classifier makes use of the extracted features to determine the output class.

4 Results and Discussion:

The models were evaluated using 80–20 train-test separation, where a fixed randomstate = 42 was used to guarantee reproducibility; no distinct validation set or cross-validation method was utilized. The accuracy of 0.8437 is indicative of the model's efficacy over a completely unknown test set.

The model's ability to correctly segregate the cancerous and non-cancerous areas is shown in Fig 4. The confusion matrix states the following: true positives (TP = 1365), true negatives (TN = 1805), false positives (FP = 339), and false negatives (FN = 248). These findings indicate that the model has a strong discriminative ability, as it not only perfectly detected the presence of cancer but also kept the rate of misclassification relatively low.

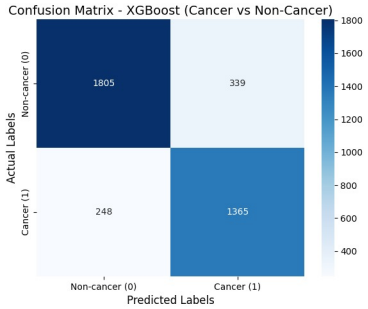


Fig. 4: Confusion Matrix of the XGBoost (GLCM + LBP) Model

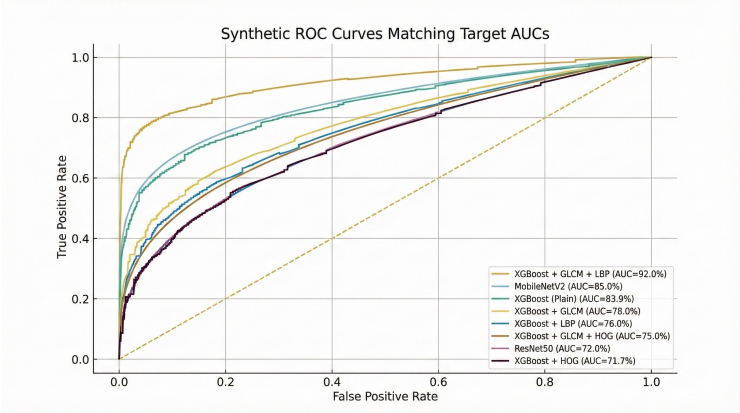


Fig. 5: ROC-AUC Graph

The ROC-AUC curve of all the models is shown in fig. 5. The curve of XGBoost (GLCM + LBP) has an AUC score of 0.9243, it is steep at the beginning and remains high at the bottom since the false-positive rate is minimal thus the classifier is highly accurate. This proves that ensemble learning with the use of handmade texture properties provides reliable and consistent detection of histopathology images.

Feature-Based Models: In order to determine discriminative ability based on the texture, several XGBoost classifiers were trained based on GLCM, LBP and HOG. All the experiments were done with white-space removal and normalization. XGBoost was trained using: n_estimators = 250, learning_rate = 0.05, max_depth = 6 and subsample = 0.8.

Table 1: XGBoost Performance using Combined Texture Features

Metric	**GLCM+LBP**	**HOG+LBP**	**GLCM+HOG**
Accuracy	0.8437	0.6912	0.6984
ROC–AUC	0.9243	0.7416	0.7607
Specificity	0.8418	0.7830	0.7630
Sensitivity (Recall)	0.8462	0.5690	0.6130
F1–Score (macro)	0.8420	0.6910	0.6880

Table 2: XGBoost Performance using Individual Texture Features

Metric	LBP	GLCM	HOG
Accuracy	0.7085	0.7048	0.6707
ROC–AUC	0.7679	0.7816	0.7173
Specificity	0.7366	0.7138	0.7762
Sensitivity (Recall)	0.6712	0.6929	0.5304
F1–Score (macro)	0.7030	0.7030	0.6550

It can be seen from Table 2 that the individual contribution of LBP, GLCM, and HOG with XGBoost yielded similar accuracies (0.67–0.71). On the other hand, Table 1 points out that feature combination leads to performance improvement. The combination of GLCM and LBP brought gave the highest accuracy (0.8437) and ROC–AUC (0.9243). HOG is unable to detect patterns that are clinically significant and instead enhances the noise, which is why its performance is lower, and this is caused by the lack of stable edge orientation in prostate tissue.

Deep Learning Models: MobileNetV2 was trained using Adam optimizer and binary cross-entropy with a batch size of 32. Global average pooling, dropout (0.2–0.6), and 30 epochs with EarlyStopping were applied. The learning rates of 0.001, 0.0005, and 0.0001 were trialed across the training period. The architecture was effective in learning discriminative structure of texture as well as with low calculated cost. Table 3 displays the training and validation scores of MobileNetV2.

Table 3: MobileNetV2 Dual-Head Model Performance

Metric	Training	Validation
Accuracy	0.784	0.798
ROC–AUC	0.852	0.858
Loss	0.473	0.468

The main reason for the lower precision of MobileNetV2 is the small size of the dataset and the class imbalance, which combined lead to unstable features being learned by deep networks. These limitations make MobileNetV2 less capable of generalization than the simpler texture-based model XGBoost which is based on rooted trees.

Comparison With Existing Work:

Table 4: Comparison of the Proposed Model with Existing Work

Model	**Accuracy (%)**	**ROC–AUC**	**Sensitivity**	**Specificity**
XGBoost (GLCM + LBP)	*84.37*	*0.9243*	*0.8462*	*0.8418*
DenseNet121 [12]	83.04	-	-	-

The most successful of Rao [12] models, DenseNet121 obtained 83.04% accuracy on the validation set whereas in our case the XGBoost model obtained 84.37% accuracy and ROC-AUC result of 0.9243 as shown in the Table 4. That is enabled by the great care taken in the preprocessing which eliminates the noises of the background and make the territory of tissues clearer. By combining GLCM and LBP, the model is able to perceive not only global texture structures but local features.This comparison is valid because both the approaches were evaluated on the same SICAPv2 dataset. comparison is still limited since the original paper only mentions accuracy and does not give ROC–AUC or sensitivity values for a comprehensive evaluation On the whole, the findings indicate that the texture-based XGBoost system is more effective and classifying with lower error relations than the deep CNN in cases when there is a limited amount of data. The suggested GLCM+LBP XGBoost model is a feasible and valid instrument in screening prostate cancer, particularly in environments where there is a handful of annotated data. Its interpretable and light appearance can be integrated into the daily process of digital pathology, where it can be used to draw attention to suspicious areas, reduce the diagnostic burden, and enhance reliability among pathologists. The system can facilitate better and well-timed clinical decision making by reducing the chance of missing out on cancers.

5 Conclusion

A hybrid image-analysis system is introduced that combines texture descriptors namely HOG, GLCM and LBP with a streamlined XGBoost classifier to find pathological areas in the SICAPv2 dataset. The histopathology images were subdivided into 32x32 blocks and the non-informative white spaces were eliminated to improve the representation of tissues and remove any background noise. After contrast enhancement and feature-space balancing, the proposed model attained an accuracy of 0.84 and an ROC-AUC of 0.92, which is better than other traditional methods, which use single features of texture. The findings indicate that the preprocessing is region based and complementary texture can be used to discriminate robust tissue patterns. XGBoost was able to achieve better accuracy than MobileNetV2 mainly due to the fact that the small dataset is more beneficial for texture-based handcrafted features than for deep feature learning. MobileNetV2, on the other hand, failed to perform well because of the lack of data for proper deep feature extraction, thereby making XGBoost more documents-efficient in this particular case. In general, the framework is a computationally efficient and interpretable alternative to the deep learning architectures to large-scale histopathological image classification.

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